



Your Thesis Title Here

A thesis submitted in partial fulfillment of
requirements for the award of the degree of

Master of Science (MS)

by

Your Name

(I2MS345)

5th Year BS-MS

Department of Physical Sciences

Indian Institute of Science Education and Research Kolkata, India

Supervised by

Prof. Supervisor Name

Indian Institute of Science Education and Research Kolkata, India

Co-supervised by

Prof. Co-Supervisor Name

Affiliation

Coordinated by

Prof. Coordinator Name

Indian Institute of Science Education and Research Kolkata, India

September 17, 2026

Declaration

I, Your Name, roll number 12MS345, a student of the Department of Physical Sciences of the 5th Year BS-MS program of Indian Institute of Science Education and Research Kolkata, India, hereby declare that this thesis is my own work and, to the best of my knowledge, it neither contains materials previously published or written by any other person, nor has it been submitted for any degree/diploma or any other academic award anywhere before. I have used the originality checking service to prevent inappropriate copying.

I also declare that all copyrighted material incorporated into this thesis is in compliance with the Indian Copyright Act, 1957 (amended in 2012) and that I have received written permission from the copyright owners for my use of their work.

I hereby grant permission to Indian Institute of Science Education and Research Kolkata, India to store the thesis in a database which can be accessed by others.

Date: September 17, 2026

Place: IISER Kolkata, Mohanpur

.....
(Signature of author)

Your Name

Indian Institute of Science Education and Research Kolkata, India

Certificate

This is to certify that the thesis titled “Your Thesis Title Here” submitted by Your Name, a student of the Department of Physical Sciences of the 5th Year BS-MS program of Indian Institute of Science Education and Research Kolkata, India, is based upon his own research work under our supervision. We also certify, to the best of our knowledge, that neither the thesis nor any part of it has been submitted for any degree/diploma or any other academic award anywhere before.

.....

(Signature of supervisor)

Prof. Supervisor Name

Indian Institute of Science Education and
Research Kolkata, India

Date: September 17, 2026

Place: IISER Kolkata, Mohanpur

.....

(Signature of co-supervisor)

Prof. Co-Supervisor Name

Affiliation

Date: September 17, 2026

Place: IISER Kolkata, Mohanpur

Acknowledgements

Let's just say, I've never been the best at predicting how much effort something will require...

Contents

Declaration	ii
Certificate	iii
Acknowledgements	iv
List of Figures	vii
List of Tables	viii
List of Listings	ix
List of Symbols and Abbreviations	x
Todo list	xi
Abstract	xii
1 Getting Started	1
1.1 Project Structure	1
1.2 Metadata Configuration	1
1.3 Compilation	2
1.4 Class Options and Toggles	2
2 Features and Customization	3
2.1 Font Customization	3
2.2 Theorem Environments	3
2.3 Mathematics and Physics	4
2.4 Citations and Bibliography	4
2.5 Figures and Tables	5
2.5.1 Figures and Subfigures	5
2.5.2 Tables and Subtables	6
2.5.3 Context-Based Capitalization Rule	8

2.6	Code Listings	8
2.7	To-Do Notes	9
A	First Appendix	10
A.1	Supplementary Information	10
	Bibliography	11

List of Figures

2.1	The official emblem of IISER Kolkata as an example of a single standalone figure. . . .	5
2.2	Multi-panel figure example: comparing institutional insignia side-by-side using the <code>subfigure</code> environment.	6

List of Tables

2.1	Fundamental physical constants in SI units.	7
2.2	Multi-table comparison of kinematic and dynamic physical quantities.	7
2.3	Context-based capitalization commands provided by <code>cleveref</code>	8
A.1	Supplementary detector calibration parameters.	10

List of Listings

2.1	A simple Python script demonstrating the minted configuration.	8
-----	--	---

List of Symbols and Abbreviations

$ \Psi\rangle$	Quantum state vector
c	Speed of light in a vacuum

Todo list

	This is a margin note. It automatically fits the page margin.	9
	This is an inline todo note, useful for longer comments or reminders inside the text body. . . .	9

Abstract

This thesis explores a novel approach to constructing a finite-dimensional holographic scalar field theory.



Getting Started

Welcome to the IISER Kolkata MS and PhD thesis template. This document serves as the official documentation, demonstrating how to use the template effectively to produce a beautiful, correctly formatted thesis.

I.1 Project Structure

The template is divided into several modular components to keep your workspace clean:

- `main.tex`: The root document where you configure metadata and include other files.
- `IISERKthesis.cls`: The core LaTeX class file that manages page layout, margins, headers, and frontmatter.
- `IISERKphysics.sty`: A custom package that loads mathematics tools, defines theorem environments, and configures code blocks.
- `frontmatter/`: Contains individual files for your abstract, acknowledgements, certificate, and declaration.
- `chapters/` and `appendices/`: Directories to house your main textual content.
- `references.bib`: BibLaTeX database containing your literature citations.

I.2 Metadata Configuration

In `main.tex`, you will find several macros to define your personal details. These automatically populate the title page, declaration, and certificate:

```
1 \title{Your Thesis Title Here}
2 \author{Your Name}
3 \rollno{12MS345}
```

```

4 \program{5th Year BS-MS}
5 \department{Department of Physical Sciences}
6 \supervisor{Prof. Supervisor Name}

```

1.3 Compilation

This template relies on `biber` for bibliography management and `minted` for syntax-highlighted code blocks. Because of `minted`, you *must* compile with the `-shell-escape` flag.

The easiest way to compile is using the included `.latexmkrc` configuration:

```

1 latexmk main.tex

```

This automates multiple compilation passes and stores all auxiliary files (like `.aux` and `.log`) neatly in a hidden `.build_tex/` folder.

1.4 Class Options and Toggles

The template automatically generates several lists at the beginning of the document (Table of Contents, List of Figures, List of Tables, List of Listings, Nomenclature, and a Todo list).

You can easily disable any of these by passing toggles to the document class in `main.tex`:

```

1 % Disables the Todo list and List of Tables
2 \documentclass[notodo, nolot]{IISERKthesis}

```

Available toggles include:

- `notodo`: Completely disables the `todonotes` package and removes the Todo List. Ensure you use this flag for your final submission!
- `nolof`: Removes the List of Figures.
- `nolot`: Removes the List of Tables.
- `nolol`: Removes the List of Listings.
- `nonom`: Removes the Nomenclature (List of Symbols).

2

Features and Customization

This chapter highlights the advanced typographical features and custom environments built into the `IISERKphysics.sty` package.

2.1 Font Customization

You can easily switch out the document fonts by using the font macros in `main.tex`. These macros take an optional LaTeX package name and a styling command.

```
1 % Example: Use 'tgheros' for sans-serif headings, 'tgtermes' for Times main text
2 \TitleFont[tgheros]{\sffamily}
3 \TextFont[tgtermes]{\rmfamily}
4 \MathFont{newtxmath}
```

If you leave these commented out, the template elegantly defaults to Garamond.

2.2 Theorem Environments

We provide beautifully styled theorem environments powered by `tcolorbox`. These boxes are fully integrated with `cleveref` for smart cross-referencing.

Theorem 2.1 (No-Hair Theorem)

Stationary black holes are characterized entirely by mass M , charge Q , and angular momentum J .

Lemma 2.2

This is a lemma without a title. Notice how the numbering follows the theorem counter chapter-wise.

Definition 2.3 (Event Horizon)

The boundary defining the region of space from which nothing can escape.

Remark 2.4 (Historical Note)

The term was popularized by John Wheeler in 1967.

You can reference these dynamically: use `\cref` for lowercase mid-sentence citations (`theorem 2.1` and `definition 2.3`) and `\Cref` when opening a sentence (`Theorem 2.1` characterizes black hole geometries).

2.3 Mathematics and Physics

Essential math packages like `amsmath`, `mathtools`, `slashed`, and `tensor` are preloaded. The `physics` package provides excellent shorthand for bra-ket notation and derivatives:

$$\hat{H} |\Psi\rangle = i\hbar \frac{\partial}{\partial t} |\Psi\rangle \quad (2.1)$$

Equation (2.1) is the time-dependent Schrödinger equation, which can also be cited mid-sentence as equation (2.1).

You can also log symbols to the Nomenclature automatically. For instance, consider the speed of light c .

2.4 Citations and Bibliography

Bibliography management is configured out of the box using `biblatex` with the `biber` backend. Citations are stored in `references.bib` (registered via `\addbibresource{references.bib}` in `main.tex`), and the bibliography is printed at the end of the document with `\printbibliography[heading=bibintoc]`.

The template provides an author-year citation style (`citestyle=authoryear`) paired with a numeric bibliography list (`bibstyle=numeric`). We provide a custom implementation of `\textcite` that links both the author name and the publication year directly to the bibliography entry in dark green (`citecolor`), while keeping parentheses unlinked.

- **Narrative citation:** Use `\textcite{key}` when the author's name forms a natural part of the sentence:

As formulated by Einstein (1915), gravitational interactions arise from the curvature of spacetime.

- **Parenthetical citation:** Use `\parencite{key}` when referring to literature parenthetically:

Gravitational radiation from black hole collisions was analyzed in seminal work (Hawking 1971), extending theorems on gravitational collapse (Penrose 1965).

- **Multiple citations:** Separate multiple citation keys with commas inside `\parencite{...}`:

Foundational developments in general relativity and singularity formation are well documented (Hawking 1971; Penrose 1965; Einstein 1915).

Here is how you write these in your source:

```
1 % Narrative citation: Einstein (1915)
2 As formulated by \textcite{einstein1915}, gravitational interactions...
3
4 % Parenthetical citation: (Hawking 1971)
5 Gravitational radiation was analyzed in seminal work \parencite{hawking1971}...
6
7 % Multiple references: (Hawking 1971; Penrose 1965)
8 Foundational developments are well documented \parencite{hawking1971,
  ↪ penrose1965}.
```

2.5 Figures and Tables

The template provides first-class support for single and composite graphics via `graphicx` and `subcaption`, as well as publication-grade tables via `booktabs`. All figures and tables integrate seamlessly with `cleveref` for automatic, contextual referencing.

2.5.1 Figures and Subfigures

Use the standard `figure` environment for individual graphics, and the `subfigure` environment (provided by `subcaption`) for multi-panel composite figures.

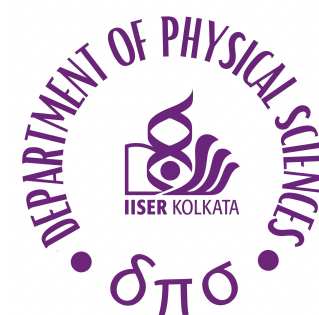
Figure 2.1 displays a standalone figure at the beginning of a sentence. In contrast, mid-sentence citations adopt the lowercase form (for example, as illustrated in **figure 2.1**). When presenting related data or comparisons side-by-side, use subfigures as shown in **figure 2.2**. **Figure 2.2** groups two institutional insignia under a single caption. You can reference individual panels as **figure 2.2a** and **figure 2.2b**, cite a combined range as **figures 2.2a** and **2.2b**, or begin a sentence with **Figure 2.2a**.



Figure 2.1: The official emblem of IISER Kolkata as an example of a single standalone figure.



(a) IISER Kolkata emblem.



(b) Department of Physical Sciences.

Figure 2.2: Multi-panel figure example: comparing institutional insignia side-by-side using the `subfigure` environment.

Here is the source code for constructing composite subfigures:

```

1 \begin{figure}[htpb]
2   \centering
3   \begin{subfigure}[b]{0.45\textwidth}
4     \centering
5     \includegraphics[width=\textwidth]{panel_a.png}
6     \caption{First observation panel.}
7     \label{fig:panel_a}
8   \end{subfigure}
9   \hfill
10  \begin{subfigure}[b]{0.45\textwidth}
11    \centering
12    \includegraphics[width=\textwidth]{panel_b.png}
13    \caption{Second observation panel.}
14    \label{fig:panel_b}
15  \end{subfigure}
16  \caption{Side-by-side composite figure with shared caption.}
17  \label{fig:composite}
18 \end{figure}

```

2.5.2 Tables and Subtables

Avoid vertical lines in scientific tables. The `booktabs` package provides clean horizontal rules: `\toprule`, `\midrule`, and `\bottomrule`. Place the `\caption` *above* the table body, adhering to standard scientific publishing conventions.

Table 2.1 demonstrates a standard single table. Mid-sentence citations automatically take the lower-case form (for example, as summarized in table 2.1). For side-by-side comparison tables, the `subtable` environment groups multiple tables under a common label, as shown in table 2.2. Individual subtables can be cited mid-sentence as table 2.2a and table 2.2b, or together as tables 2.2a and 2.2b. At the start of a

sentence, use `\Cref` (for instance, “[Table 2.2a](#) presents kinematics”).

Table 2.1: Fundamental physical constants in SI units.

Quantity	Symbol	Value	Unit
Speed of light in vacuum	c	2.9979×10^8	m s^{-1}
Planck constant	h	6.6261×10^{-34}	J s
Gravitational constant	G	6.6743×10^{-11}	$\text{m}^3 \text{kg}^{-1} \text{s}^{-2}$
Elementary charge	e	1.6022×10^{-19}	C

Table 2.2: Multi-table comparison of kinematic and dynamic physical quantities.

(a) Kinematic parameters.			(b) Dynamic parameters.		
Variable	Symbol	SI Unit	Variable	Symbol	SI Unit
Position	$\vec{x}$	m	Mass	m	kg
Velocity	$\vec{v}$	m s^{-1}	Momentum	$\vec{p}$	kg m s^{-1}
Acceleration	$\vec{a}$	m s^{-2}	Force	$\vec{F}$	N

Here is how you format multiple side-by-side subtables in your code:

```

1 \begin{table}[htpb]
2   \centering
3   \caption{Comparison of two related parameter sets.}
4   \label{tab:comparison}
5   \begin{subtable}[b]{0.45\textwidth}
6     \centering
7     \caption{First parameter set.}
8     \label{tab:part_a}
9     \begin{tabular}{lc}
10      \toprule Parameter & Value \\ \midrule A & 1.0 \\ \bottomrule
11    \end{tabular}
12   \end{subtable}
13   \hfill
14   \begin{subtable}[b]{0.45\textwidth}
15     \centering
16     \caption{Second parameter set.}
17     \label{tab:part_b}
18     \begin{tabular}{lc}
19      \toprule Parameter & Value \\ \midrule B & 2.0 \\ \bottomrule
20    \end{tabular}
21   \end{subtable}
22 \end{table}

```

2.5.3 Context-Based Capitalization Rule

Scientific publishing standards (such as APS, IEEE, and Chicago Manual of Style) prescribe *context-based capitalization* for cross-references:

- **Sentence-initial references** (`\Cref`): When opening a sentence, use `\Cref{...}`. It automatically capitalizes the output (e.g., `\Cref{tab:constants}` yields “Table 2.1”).
- **Mid-sentence references** (`\cref`): Within running text or parentheses, use `\cref{...}`. It automatically preserves lowercase (e.g., `\cref{tab:constants}` yields “table 2.1”).

Table 2.3 summarizes the context-based output across primary environments.

Table 2.3: Context-based capitalization commands provided by <code>cleveref</code> .		
Environment	Sentence Start (<code>\Cref</code>)	Mid-Sentence (<code>\cref</code>)
Figure	<code>\Cref{fig:...}</code> → Figure 2.1	<code>\cref{fig:...}</code> → figure 2.1
Table	<code>\Cref{tab:...}</code> → Table 2.1	<code>\cref{tab:...}</code> → table 2.1
Equation	<code>\Cref{eq:...}</code> → Equation (2.1)	<code>\cref{eq:...}</code> → equation (2.1)
Theorem	<code>\Cref{thm:...}</code> → Theorem 2.1	<code>\cref{thm:...}</code> → theorem 2.1
Section	<code>\Cref{sec:...}</code> → Section 2.4	<code>\cref{sec:...}</code> → section 2.4

2.6 Code Listings

The template utilizes `minted` for professional code blocks as well as inline code. Crucially, we use the `accsupp` package to render line numbers uncopyable, so readers can effortlessly copy your code from the PDF without grabbing the line numbers.

For inline code, use `\mintinline{<language>}{<code>}` or the shorthand `\code{<code>}`. Inline code is configured with automatic line breaking (`breaklines=true`, `breakbytoken=true`) so that long file paths, commands, or code tokens wrap automatically according to the text width without overflowing into page margins.

```

1 import numpy as np
2
3 def calculate_energy(mass):
4     """Calculates rest energy E = mc^2"""
5     c = 299792458 # m/s
6     return mass * c**2

```

Listing 2.1: A simple Python script demonstrating the `minted` configuration.

2.7 To-Do Notes

As you draft your thesis, you can leave yourself notes using the `\todo{}` command.

This is an inline todo note, useful for longer comments or reminders inside the text body.

These notes compile into a handy index at the front of the document. When you are ready to finalize your thesis, simply add the `notodo` class option (see [section 1.4](#)) to hide all notes and the list simultaneously.

This is a margin note. It automatically fits the page margin.



First Appendix

This is an example of an appendix chapter. Appendices automatically use alphabetic lettering (Appendix A, Appendix B) instead of standard numeric chapters.

A.1 Supplementary Information

You can include extra derivations, long tables of data, or additional code snippets here that are relevant but too detailed for the main text.

Table A.1 lists supplementary calibration parameters at the beginning of a sentence. Notice how opening a sentence with `\Cref` produces the capitalized form (**Table A.1**), while mid-sentence references such as **table A.1** preserve lowercase, with both adopting the appendix letter prefix A . 1.

Table A.1: Supplementary detector calibration parameters.

Receiver Channel	Center Frequency (MHz)	Gain (dB)	Noise Temp (K)
CH1 (L-Band)	1420.4	45.2	32.1
CH2 (OH-Line)	1665.4	44.8	34.5
CH3 (OH-Line)	1667.3	45.0	33.8

Bibliography

- [1] Albert Einstein. “The Field Equations of Gravitation”. In: *Sitzungsberichte der Preussischen Akademie der Wissenschaften* 1 (1915), pp. 844–847.
- [2] Stephen W Hawking. “Gravitational radiation from colliding black holes”. In: *Physical Review Letters* 26.21 (1971), p. 1344. DOI: [10.1103/PhysRevLett.26.1344](https://doi.org/10.1103/PhysRevLett.26.1344).
- [3] Roger Penrose. “Gravitational collapse and space-time singularities”. In: *Physical Review Letters* 14.3 (1965), p. 57. DOI: [10.1103/PhysRevLett.14.57](https://doi.org/10.1103/PhysRevLett.14.57). arXiv: [1401.1234](https://arxiv.org/abs/1401.1234).